QPCODE **Reg.No :.....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

4th semester MGU-UGP(Honours) Examination

DSC A- MG4DSCCFS201-APPLIED CRYPTOGRAPHY

(2024 ADMISSION ONWARDS)

Duration: 1.5Hrs **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type

Questions Answer any **10** questions Each question carries **1** marks

1. Define cryptography. [U]/[CO1]
2. List cryptographic terminologY [U]/[CO1]
3. Classify the applications of cryptography [A]/[CO3]
4. Outline private key is also called as. [An]/[CO2]
5. List cryptographic Primitives. [An]/[CO2]
6. Formulate a hash function denoted as. [C]/[CO4]
7. List the application of hash function. [C]/[CO4]
8. Define Block cipher. [An]/[CO2]
9. Write classification of block cipher mode operation. [C]/[CO4]
10. Write expansion of PKI . [A]/[CO3]
11. Define message authentication code. [An]/[CO2]
12. Write expansion of HMAC. [An]/[CO2]

[1x10 = 10]

Part B

Short Essay Type Questions

Answer any 4 questions

Each question carries 5 marks

13. Classify symmetric key cryptography vs. asymmetric key cryptography. [An]/[CO2]
14. Explain Stream ciphers and block ciphers. [An]/[CO2]
15. Explain Electronic Codebook (ECB) and Cipher Block Chaining (CBC).[C]/[CO4]
16. Explain Digital signatures [A]/[CO3]
17. Explain Diffie-Hellman Key Exchange. [A]/[3]
18. Explain Secure Hash Algorithm. [C]/[4]

[4x5 = 20]

Part C

Essay Type Questions

Answer any **2** questions

Each question carries **10** marks

19. Explain Data Encryption Standard (DES) with suitable diagram. [An]/[CO2]
20. Explain Advanced Encryption Standard (AES). [C]/[CO4]
21. Explain RSA Algorithm with an example. [A]/[CO3]

[2x10 = 20]